

SURVEY

A Systematic Review of Machine Learning and Deep Learning Techniques for Brain Tumor Classification, and Grading

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ABSTRACT Brain tumors rank among the most lethal forms of cancer, and their early detection is crucial for improving patient outcomes. Beyond timely identification, accurately determining tumor type and grade is essential for guiding treatment strategies. While Magnetic Resonance Imaging (MRI) is a mainstay in tumor detection and diagnosis, the precise classification of brain tumors remains challenging due to variations in their morphology, including shape and size. In recent years, artificial intelligence—particularly machine learning methods such as Support Vector Machines (SVM) and k-nearest neighbors (KNN), and deep learning architectures including Convolutional Neural Networks (CNNs)—has emerged as a powerful tool for classifying brain tumor types and grades from MR images. This review provides a comprehensive examination of contemporary studies that leverage these methods for tumor classification. The analysis is based on a structured search of major scientific databases and includes 81 studies published between 2014 and 2025 that met predefined inclusion criteria. Specifically, it analyzes the classification techniques employed, the datasets and their characteristics, feature extraction approaches, and performance evaluations reported in the literature. Reported results show that deep learning methods generally outperform traditional machine learning models, with many studies achieving accuracies above 90% in tumor type or grade classification tasks. In addition to summarizing existing work, this review provides a unified perspective by jointly examining tumor type and multi-class grade classification studies, highlighting understudied areas such as non-glioma grade prediction and common dataset limitations. By synthesizing existing findings, the review identifies key research gaps and proposes directions for future investigations. A key future direction highlighted in this review is the need for more comprehensive multi-class tumor grade prediction studies, particularly for non-glioma tumor types. The insights presented here aim to aid researchers and clinicians in developing more accurate and efficient brain tumor classification systems, leading to earlier diagnosis, reduced diagnostic variability, and more informed treatment planning in clinical practice.

INDEX TERMS Brain tumor, deep learning, machine learning, magnetic resonance imaging.

I. INTRODUCTION

The brain is the organ that controls the activities of the human body and serves as the central nervous system's management center. "Tumor" is a general term that refers to abnormal tissue growth resulting from the uncontrolled proliferation of cells. Likewise, a "brain tumor" refers to

the unchecked and abnormal growth and division of cells in the brain [1]. Brain tumors comprise a heterogeneous group of intracranial neoplasms that may originate from the brain itself (primary tumors such as gliomas and meningiomas) or spread from cancers elsewhere in the body (metastatic tumors). Depending on their type, grade, and location, many of these tumors can be life-threatening and may severely impair central nervous system function. Early diagnosis is critical because tumor grade and extent of infiltration

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strongly influence treatment options and prognosis; low-grade tumors may be managed surgically with favorable outcomes, whereas delayed identification of high-grade or rapidly progressing tumors often leads to limited therapeutic success and poorer survival rates. This pathological condition is defined as an abnormally proliferating cluster of cells that can severely damage the central nervous system. These intracranial lesions are categorized as either benign or malignant. Typically, benign neoplasms remain localized and do not invade surrounding tissues, but if they put pressure on important structures, they may require surgical intervention [2]. Malignant neoplasms, on the other hand, can proliferate and spread to surrounding tissues. Many factors have led to the adoption of different classification strategies for these conditions. Classification is a challenging task due to the complex biology and diversity of the nervous system. The World Health Organization (WHO) has conducted studies over the years on the classification of brain tumors. Currently, the WHO classification of brain tumors introduced in 2021 is in use. Treatment options vary depending on the lesion's location, size, and classification. A correct diagnosis is essential for successful treatment. Therefore, it is crucial to detect the tumor at an early stage and accurately classify its type. Tumor grades are established based on histological criteria. The WHO grading system, which rates tumors from I to IV, is the most widely used system for grading brain tumors [3]. The use of radiological imaging is a common technique for classifying tumor types. Recently, various machine learning (ML) and deep learning (DL) approaches have been proposed by researchers to classify tumor types using radiological data [4]. Studies have shown that ML and DL techniques in radiology have achieved remarkable performance. Moreover, recent research indicates that decision support systems can automatically assess tumor grades by leveraging these techniques.

Artificial intelligence (AI), particularly ML and DL, has enabled automated analysis of complex medical images. ML algorithms learn discriminative patterns from labeled data, whereas DL architectures such as convolutional neural networks (CNNs) can extract high-level features directly from MRI without manual feature engineering. Thanks to these capabilities, AI-based systems have become increasingly effective for brain tumor classification and grading tasks [5], [6].

In recent years, AI-based approaches have begun to transform brain tumor analysis in practical and clinically meaningful ways. DL models can automatically identify subtle radiological patterns that may be difficult for human observers to distinguish, enabling faster and more consistent tumor type classification. Furthermore, AI systems assist in estimating tumor grade by analyzing features related to cellularity, necrosis, and infiltration on MRI, which supports radiologists in making more accurate and reproducible assessments. These technological advances not only reduce diagnostic variability but also have the potential to shorten the

time to treatment planning, ultimately improving workflow efficiency in neuro-oncology.

Despite these strengths, several factors still complicate brain tumor classification. These include heterogeneous tumor morphology, overlapping radiological characteristics between grades, and substantial inter-observer variability among radiologists. Additionally, conventional manual interpretation is often subjective and may overlook subtle textural or spatial patterns in MRI. Recent AI-based approaches, however, have shown considerable advantages over traditional methods. For example, DL models have achieved high performance in multi-class tumor classification and grade prediction, outperforming classical ML and manual radiological assessment in several studies [7]. These works demonstrate that AI systems can extract discriminative features that are difficult to perceive visually, leading to improved diagnostic accuracy, reproducibility, and clinical decision-making.

Different tumor types and their underlying histo-molecular features directly influence the complexity of the classification task, the choice of imaging modalities, and the performance of AI-based models. Therefore, an overview of major brain tumor categories is provided below to establish the clinical and biological context for subsequent AI-driven analyses.

A. BRAIN TUMOR TYPES

Brain tumors are classified according to various classification criteria. Classification schemes are based on many factors, such as histological features, cell type of origin, location, genetic and molecular features. WHO has published classification systems for the classification of central nervous system tumors and updated them over the years. The WHO 2021 classification, which adopts the latest histo-molecular classification system, has been published. Figure 1 provides a summary of this classification.

While tumor type classification focuses on identifying the origin, histological characteristics, and molecular profile of each lesion, grading provides complementary information about the biological aggressiveness and expected clinical behavior of these neoplasms [8].

B. GRADES OF BRAIN TUMORS

Histological criteria are used to determine the grades of tumors. The WHO grading system, which classifies brain tumors into grades ranging from I to IV, is the most widely used framework for tumor grading [3]. Grade I, also known as benign tumors, typically exhibit a slow growth pattern and rarely invade the surrounding tissues. They often have well-defined borders, and cells appear relatively normal under a microscope. Grade II (Low-Grade) tumors are still considered slow-growing, but they may show more abnormal cell characteristics. They may spread to nearby tissues to some extent. Grade III (Anaplastic) tumors are considered malignant and are faster-growing. Cells appear more abnormal under a microscope, and the tumor is more

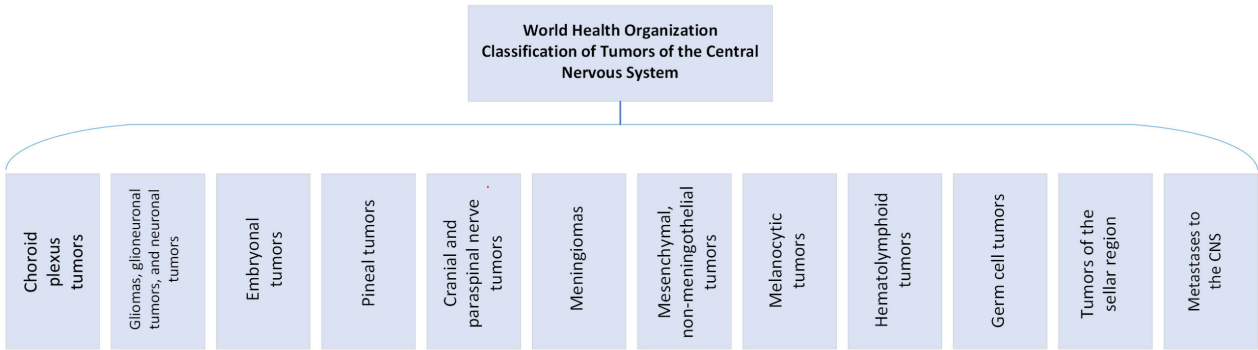


FIGURE 1. A summary of the World Health Organization (WHO) 2021 classification of central nervous system tumors.

likely to invade surrounding tissues. Grade IV (Glioblastoma Multiforme) tumors are the most aggressive and malignant tumors. They grow rapidly and invade nearby tissues. Cells are highly abnormal and may have areas of dead cells (necrosis). Glioblastoma multiforme is an example of a Grade IV brain tumor. Traditionally, this procedure is conducted through histological inspection. However, new articles have demonstrated the feasibility of performing it using magnetic resonance imaging (MRI), making it a non-invasive approach. Figure 2 provides a summary of this grading.

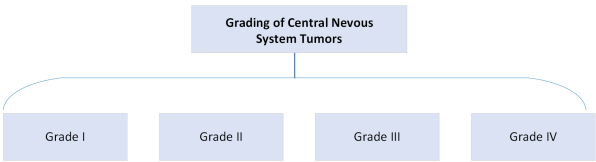


FIGURE 2. WHO grading system for central nervous system tumors.

C. BRAIN IMAGING TECHNIQUES

Computed tomography (CT), MRI, positron emission tomography (PET), and blood and urine tests are used to diagnose brain tumors. In cases of brain tumors, imaging techniques are often preferred instead of biopsies, as the biopsy process can be time-consuming when confirming the tumor’s type and grade. The most commonly used imaging methods are MRI, PET, and CT [9], [10].

PET scans are nuclear imaging techniques. By injecting a radioactive tracer, usually fluorodeoxyglucose (FDG), the targeted organ can be imaged in three dimensions. The tracers attach to compounds such as glucose. Since brain and cancer cells retain glucose, the affected area can be observed more clearly during the scan. However, high-grade tumors cannot always be distinguished on FDG-PET images [11].

CT scans provide detailed analyses. Images are obtained using X-rays, which emit high levels of radiation. CT scans are generally not recommended for intensive examinations, as repeated scans significantly increase radiation exposure. In such cases, MRI is usually preferred [11], [12].

MRI is a diagnostic modality that utilizes potent magnetic fields and radiofrequency signals to produce high-resolution

pictures of inside tissues and organs. MRI is particularly effective for detecting brain malignancies. Brain tumors may exhibit various characteristics, including iso-intensity (where tumor tissues show similar signals to surrounding tissue), hypointensity (where regions appear darker than tumor tissue), hyperintensity (regions with increased signal intensity), tumor heterogeneity, and tumor homogeneity [13]. Ambiguities in the imaging process may also arise due to conditions such as perilesional edema in neighboring tissues. In comparison to CT and PET, MRI provides enhanced soft-tissue contrast, avoids ionizing radiation exposure, and delivers various sequences that yield complementing structural and pathological insights. These advantages make MRI the preferred imaging modality for AI-based brain tumor classification, as it enables more accurate delineation of tumor boundaries, edema, and infiltration patterns. Consequently, most machine learning and deep learning studies in the literature rely primarily on MRI rather than CT or PET.

Due to these complexities, classifying or grading brain tumors can be challenging for medical experts [14]. Many researchers are working on developing faster and easier diagnostic methods based on MR images. MR imaging enables visualization of the brain in three distinct planes, adding depth to the images. These planes are illustrated in Figure 3 [15].

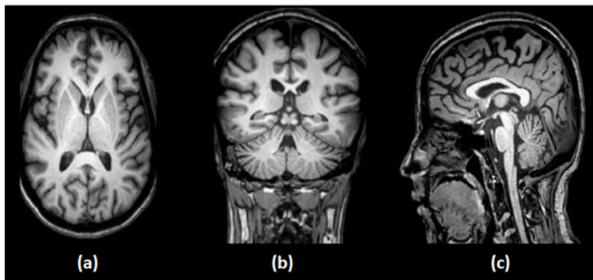


FIGURE 3. Axial (a), coronal (b), and sagittal (c) planes of brain MRI scans. Images adapted from [15] and [16].

T1-weighted MRI (T1w), T2-weighted MRI (T2w), contrast-enhanced T1-weighted MRI (ceT1w), and Fluid-Attenuated Inversion Recovery (FLAIR) sequences are widely used for the detection and classification of brain

tumors in both research and clinical practice. While some studies have found it appropriate to use a single sequence, others have preferred to work with multiple sequences. Figure 4 shows brain MRI images obtained using different sequences [16].

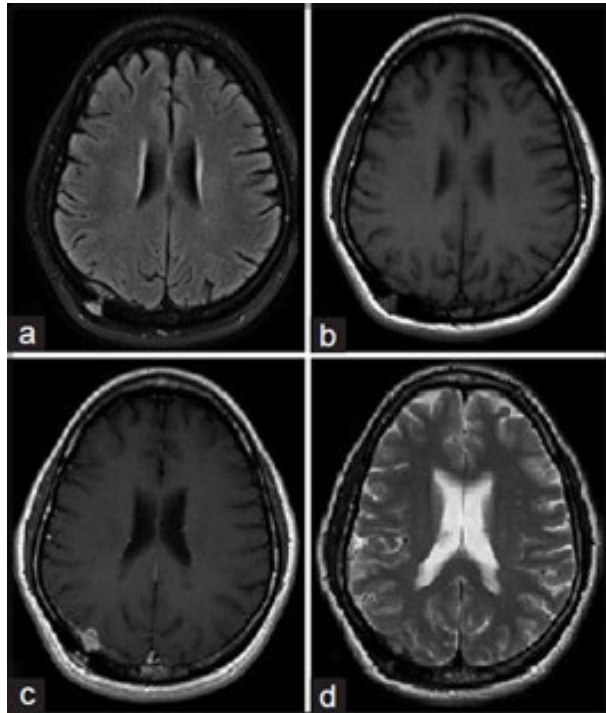


FIGURE 4. MRI modalities commonly used in brain tumor analysis: (a) FLAIR, (b) T1-weighted, (c) contrast-enhanced T1-weighted (T1ce), and (d) T2-weighted [16].

D. BRAIN TUMOR DATASETS

It has emerged as a result of the studies we have reviewed that the data required to conduct studies on brain tumors are generally obtained from open-source sources. In the literature, the data most preferred by researchers are actually BraTS, Figshare and Cancer Imaging Archive. The case numbers in these data sets are given in Table 1. All of these datasets are openly accessible for research purposes.

BraTS (Brain Tumor Segmentation) organizes an annual competition focused on brain tumor segmentation and dataset usage [17]. This competition allows researchers to test and compare their algorithms. The BraTS dataset focuses on a single type of tumor, glioma. It was first introduced in 2012 with 50 cases and has increased its sample size every year. The latest version, released in 2023, contains 2001 cases. Both high-grade and low-grade glioma cases are included, with detailed descriptions provided in Table 1.

The Figshare brain tumor dataset comprises 3,064 T1-weighted contrast-enhanced images depicting three distinct types of brain tumors [18]. It includes 233 cases, consisting of meningioma (708 slices), glioma (1,426 slices), and pituitary tumor (930 slices).

The TCGA-GBM dataset, part of The Cancer Genome Atlas (TCGA), focuses on Glioblastoma Multiforme (GBM) [19]. It contains clinical, genetic, pathological, and radiological data. GBM radiological images with 262 cases are stored in the Cancer Imaging Archive.

TABLE 1. Detail of the most preferred datasets for brain imaging studies.

Dataset Source	Dataset Number	Dataset Type
BraTS 2017	285 cases	Glioma
BraTS 2018	285 cases	Glioma
BraTS 2019	335 cases	Glioma
BraTS 2020	369 cases	Glioma
BraTS 2021	2000+ cases	Glioma
FigShare	233 cases	Meningioma–Glioma–Pituitary
GBM	262 cases	Glioblastoma Multiforme
REMBRANDT	130 cases	Glioma

The REMBRANDT dataset was originally created by the National Cancer Institute (NCI) and later transferred to Georgetown University.

While these datasets—such as BraTS, Figshare, TCGA-GBM, and REMBRANDT—are widely used in brain tumor research due to their accessibility and well-annotated nature, they also present certain limitations. One major concern is class imbalance, where some tumor types or grades are underrepresented, potentially leading to biased model performance. Additionally, most of these datasets are derived from limited sources or single centers, lacking diversity in terms of imaging devices, protocols, and patient demographics. This limited diversity may reduce the generalizability of models to real-world clinical environments.

Therefore, future research should focus on incorporating multi-center datasets and evaluating models on external, heterogeneous data to ensure broader applicability in clinical settings [20].

E. KEY CONTRIBUTION

This review summarized analyses of 81 significant articles in their fields that met the predefined inclusion criteria. Based on the reviewed literature, it has been shown that it is possible to predict the type or degree of brain tumor with computer-aided techniques. Studies conducted in recent years make it possible to classify brain tumors without surgery, thanks to AI and imaging techniques. In the article, DL and ML techniques used in the literature to distinguish and grade brain tumor type were investigated. The datasets used in the studies and parameters such as their content, accessibility, and distribution were examined. In the studies carried out, the effect of pre-process methods and other selected methods on the result was investigated. This review aims to identify which tumor types and grades have been frequently studied and to highlight areas with limited research. In doing so, it seeks to provide insights into the gaps in the current literature.

II. ARTICLE SELECTION CRITERIA

The literature search was conducted using databases such as PubMed, Scopus, and Google Scholar to ensure comprehensive coverage. Keywords such as “brain tumor,” “magnetic

resonance imaging,” “machine learning,” “deep learning,” “tumor classification,” and “tumor grading” were used in various combinations. The search focused on English-language, peer-reviewed journal articles published between 2015 and 2025. All searches were performed manually without the use of automated tools. To ensure the quality of the included studies, only articles that clearly described their methodology, datasets, and performance metrics were considered. Studies lacking sufficient methodological details, presenting unclear results, or failing to report key evaluation metrics were excluded from this review.

Studies on brain tumor analysis methods can generally be categorized into tumor detection, tumor area segmentation, and tumor classification [4]. This review specifically focuses on the classification of tumor grades within the tumor classification category.

The articles were selected based on specific criteria. This study focuses on the categorization of brain tumors or tumor grades using ML or DL methods. It exclusively includes studies that utilize MRI techniques. The use of MRI in any part of the study was considered sufficient for inclusion. Articles that employed other methodologies, such as blood analysis or demographic data, were excluded. The included studies were required to report at least one performance metric, such as accuracy, sensitivity, precision, F1-score, or the area under the receiver operating characteristic curve (AUC). Only English-language articles were considered. Review articles, conference papers, and publications limited to abstracts were excluded. Figure 3 illustrates the search strategy applied for the literature review. The search focused on studies related to brain tumors and MRI, combined with ML and DL approaches for tumor classification or grading. While the number of studies published in this field was relatively limited between 2009 and 2015, a significant increase has been observed since 2016. This growth is attributed to the widespread use of ML and DL technologies in medical imaging. In particular, the processing of brain MR images has led to innovative solutions for challenges such as tumor diagnosis, classification, and segmentation, driving increased research interest. Between 2020 and 2025, advances in DL algorithms and the availability of large datasets have further accelerated research in this area. The automatic processing of MR images for brain tumor detection and analysis has not only expedited diagnostic procedures but also improved accuracy rates. In conclusion, the graph demonstrates that brain MR image analysis using AI techniques has become a rapidly growing and high-impact research area in the field of medical imaging. The study selection process is summarized in the PRISMA 2020 flow diagram presented in Figure 6.

III. COMPREHENSIVE REVIEW

Upon reviewing the literature, significant improvements have been achieved in studies supported by AI over the past couple of decades. The merging of AI research in healthcare offers great potential for numerous scientists. Within the

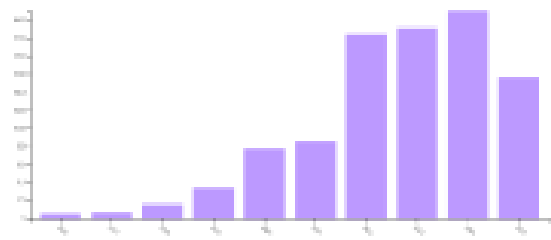


FIGURE 5. Shows annual distribution of publications related to brain tumor, MRI, and machine/deep learning.

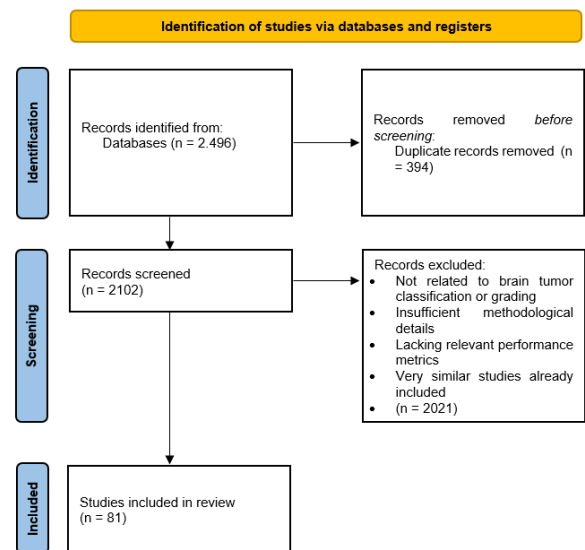


FIGURE 6. Study selection process is summarized in the PRISMA 2020 flow diagram.

domain of AI, ML and DL techniques serve as powerful tools for recognizing and categorizing brain tumors. Experts suffer from a time-consuming challenge when it comes to accurately diagnosing tumors and determining their grade. Scientists strive to create automated systems that tackle this issue. Section III-A outlines the initial procedures carried out on the data for this specific objective. The authors’ studies are examined in section III-B.

A. IMAGE PRE-PROCESSING AND FEATURE EXTRACTION TECHNIQUES

In image data, certain preprocessing steps are undertaken to make raw images suitable for analysis and processing. MRI images can be acquired from different devices, or the technology used may change over time. Such variations alter the intensity values of MRI images. Since these differences can reduce the performance of AI methods, normalization techniques are often employed. These techniques allow the image data to produce more accurate and consistent results. Normalization techniques can be of various types, such as min-max normalization, Z-score normalization, intensity normalization, etc. Models work with images of specific

fixed sizes. Therefore, images of different sizes need to be standardized or resized according to the model. This process can be done through methods such as cropping, padding, or scaling. Another pre-processing method involves the removal of non-brain regions like the skull, eyes, and nose from the brain MRI images, which is due to the nature of these images. This procedure, known as skull stripping in the literature, is performed because these regions are unnecessary for tumor classification and may reduce classification accuracy. Moreover, these areas decrease classification accuracy. Therefore, removing these regions from the image increases classification success. In brain MRI images, the image registration process is also used. This process involves transforming the images into a single coordinate system. To provide more accurate results about the tumor in patients, multiple MRI images of the patient are examined. Since these images are taken at different times and in different modalities, it is necessary to integrate the data. MRI images may contain noise due to changes in the magnetic field [21]. Filtering methods are used to eliminate noise, which can be achieved with methods like Gaussian filtering and mean filtering. Subsequently, enhancing the contrast can provide a clearer image. This ensures successful results during feature extraction and the classification phase. Feature extraction is an important step in making sense of raw data and making it usable. Raw images are difficult to analyse directly. Thanks to feature extraction, the model avoids unnecessary details and can focus on features that will be effective in classification. In brain tumors, feature extraction is often performed after the segmentation stage. Tissue features are generally used for distinguishing brains with tumors. Texture features can be derived by the application of statistical and spectral methodologies. In MRI-based brain tumor analysis, feature extraction techniques are widely used to capture tissue heterogeneity, texture characteristics, and structural patterns that are discriminative for tumor classification. The Discrete Wavelet Transform (DWT) is frequently employed because it decomposes images into multi-resolution subbands, enabling the representation of both coarse and fine-scale texture variations that are often associated with heterogeneous tumors such as gliomas and meningiomas [22], [23]. Another commonly used descriptor is the Gray Level Co-occurrence Matrix (GLCM), which models the spatial relationships between voxel intensities. By encoding contrast, homogeneity, and directional texture, GLCM effectively reflects tissue irregularities and intra-tumoral heterogeneity, properties that have been shown to correlate with tumor behavior and malignancy [24]. Local Binary Patterns (LBP) and histogram-based features also contribute to tumor characterization by capturing micro-texture transitions and the global distribution of intensity values, making them robust descriptors for differentiating tumor and non-tumor regions [25], [26]. With the increasing success of DL, many recent studies extract high-level features directly from MRI images using pretrained CNN architectures such as VGG-16, GoogleNet, AlexNet, and ResNet,

which provide richer and more abstract representations compared to handcrafted features [27], [28], [29], [30].

The performance of these pre-processing and feature extraction strategies must be evaluated using standardized metrics to ensure reliable comparison across studies.

B. PERFORMANCE EVALUATION METRICS

In the reviewed studies, several common performance metrics were used to evaluate classification models, including accuracy, sensitivity (recall), specificity, precision, and F1-score. These metrics are defined based on the confusion matrix as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (2)$$

$$Specificity = \frac{TN}{TN + FP} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (5)$$

Here, TP, TN, FP, and FN refer to true positives, true negatives, false positives, and false negatives, respectively. While accuracy is the most frequently reported metric, it may provide misleading results when class imbalance exists. This imbalance is common in brain tumor datasets, where certain tumor grades or subtypes are significantly under-represented, making accuracy an insufficient indicator of clinical performance. Therefore, sensitivity, specificity, and F1-score are critical for a more comprehensive evaluation of models, as missing a tumor case (low sensitivity) may delay diagnosis, whereas misclassifying healthy tissue as tumor (low specificity) may lead to unnecessary follow-up imaging or invasive procedures. These metrics are crucial for comparing different studies in a systematic review to ensure consistency in performance evaluation. In addition, the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curve is also widely used to assess model performance. AUC is particularly important in clinical contexts because it reflects how well a model separates tumor vs. non-tumor cases across changing decision thresholds, providing a more robust evaluation than a single accuracy value.

With these evaluation criteria established, the following sections summarize how ML and DL methods have been applied for brain tumor type classification.

C. REVIEWS ON BRAIN TUMOR TYPES CLASSIFICATION METHODS

The classification of brain tumor types by ML and DL methodologies has gained significant prominence in recent years, largely due to the enhanced accuracy and efficiency these approaches can offer in medical image analysis. Section 3.2.1 of this work, we review studies that

employ traditional ML algorithms—such as Support Vector Machines (SVM), Random Forest (RF), Decision Trees (DT), and k-Nearest Neighbors (k-NN)—to classify brain tumor types. These algorithms often rely on handcrafted feature extraction methods, where image characteristics (e.g., texture, shape, and intensity) are meticulously selected and then input into a chosen classifier. The performance of these methods can vary based on the quality of the extracted features, the dataset size, and the specific ML model employed.

In Section 3.2.2, we shift our focus to DL-based classification approaches. Unlike classical ML algorithms, DL approaches—particularly Convolutional Neural Networks (CNNs)—are highly effective in learning hierarchical features directly from imaging data without relying on handcrafted descriptors. This feature-learning capability reduces the need for manual feature engineering and often results in higher accuracy for tumor classification tasks. The studies discussed in this section typically explore architectures such as CNNs, Residual Networks (ResNet), and U-Net variants, which are adapted or fine-tuned for tumor classification. These methods are evaluated using various performance metrics, including accuracy, sensitivity, specificity, and the F1-score, offering comprehensive insights into their clinical utility.

By examining both the ML and DL-based studies side by side, this section provides a comparative understanding of how different computational methodologies can be leveraged to identify and classify brain tumors. This dual perspective also highlights the strengths and limitations of each approach and identifies the most promising directions for future research.

1) TUMOR CLASSIFICATION WITH MACHINE LEARNING

Deepak and Ameer [31] proposed a technique called deep feature fusion, which focuses on combining deep features that have been extracted from CNN models. The fused deep features are subsequently classified using SVM and k-nearest neighbors (KNN) algorithms. They also used majority voting to predict the output of CNN models trained with different loss functions. The highest accuracy values of 95.6% and balanced accuracy of 94.8% were obtained by using Majority voting on KNN models. Cheng et al. [32] explored brain tumor classification using three feature extraction strategies, namely density histograms, the gray-level co-occurrence matrix (GLCM), and the bag-of-words (BoW) model, to assess their comparative performance. The classifiers employed were SVM, k-nearest neighbors (k-NN), and k-means techniques. The SVM classifier was used to attain an accuracy of 91.14% by utilizing the features retrieved with BoW. Pattanaik et al. [25] proposed an approach for the classification of MR images of 3 kinds of most common brain tumors (glioma, meningioma, pituitary) and no-tumor. They used SVM, KNN, Naive Bayes, Decision Tree, and an ensemble classifier. The fine KNN classifier

achieved the highest performance in classifying tumor types, with an accuracy of 91.1% and an AUC of 0.96. In the study by Arakeri and Reddy [33], a computer-aided diagnosis system was designed to classify brain tumors into benign or malignant categories using MRI. After examining feature selection approaches such as PCA, ICA, and GA, the authors selected ICA. The CAD model utilized an ensemble of SVM, ANN, and k-NN classifiers tested on both T1- and T2-weighted MRI sequences. Vidyarthi et al. [34] focused on differentiating several high-grade malignant brain tumor types—Central Neurocytoma (CNC), Glioblastoma Multiforme (GBM), Gliomas (GLI), Intra Ventricular Malignant (IVMM), Metastasis (MTS)—as well as normal brain regions (NR). To accomplish this, they introduced a feature selection strategy called the Cumulative Variance Method (CVM) and applied classifiers including SVM, KNN, and neural networks. Reported classification accuracies were 88.43% for KNN, 92.5% for mSVM, and 95.86% for NN. Garg and Gard [35] proposed a hybrid ensemble classifier to classify brain tumors as benign and malignant. Authors segmented regions using Otsu's Threshold method. They used hybrid ensemble classifiers including KNN, RF, and DT based on majority voting method. Chen et al. [36] implemented an extended Kalman filter with SVM to classify tumors as benign or malignant. Celik and Inik [37] developed a novel hybrid model combining DL and optimized ML algorithms for multi-class brain tumor classification. They proposed a new CNN architecture for feature extraction and used Bayesian-optimized ML classifiers (SVM, KNN, DT, NB) for classification. The model was tested on a large MRI dataset consisting of four classes (glioma, meningioma, pituitary, and no tumor), achieving 97.15% accuracy with the CNN-KNN hybrid and 97.93% with EfficientNetB0-SVM. The proposed model outperformed other state-of-the-art CNN models in terms of both classification accuracy and time efficiency, highlighting the effectiveness of combining optimized ML algorithms with CNN-based feature extraction. Qureshi et al. [38] introduced an Intelligent Ultra-Light Brain Tumor Detection (UL-BTD) framework that combines deep features extracted from a custom Ultra-Light DL Architecture (UL-DLA) with GLCM-based textural features to form a hybrid feature space. These fused features were classified using traditional machine-learning algorithms, including SVM, k-NN, and Random Forest. Their best performance was achieved using SVM with a third-order polynomial kernel, yielding 99.23% accuracy and an F-measure of 0.99 for glioma, meningioma, and pituitary tumor detection on a CE-MRI dataset. The proposed model outperformed existing lightweight architectures by approximately 2% and required only 11.69 ms per image, demonstrating strong potential for real-time clinical and intraoperative use.

Overall, the reviewed ML studies show that classical classifiers—particularly SVM, KNN, and their ensemble combinations—provide strong and consistent performance, especially on private or well-structured datasets. In these controlled settings, ML models frequently reach 98–99%

TABLE 2. Tumor classification with machine learning.

Paper No	Data			Detection Methods	Metrics
	Source	Patient Number	Tumor Type		
[31]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	SVM k-NN	Acc=95.60
[32]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	SVM	Acc=91.14
[25]	Figshare	2870 images	822-Meningioma 826-Glioma 827-Pituitary 395-No tumor	k-NN	Acc=91.0
[33]	Private	550	280-Benign 270-Malign	SVM ANN k-NN NN	Acc=99.09 Sens=100 Spec=98.21 Acc=95.86
[34]	Private	110 patients, 660 images	133-CNC 160-GBM 155-GLI 152-IVMM 160-MTS		
[35]	TCGA-GBM	2556 samples	1278-Benign 1278-Malignant	RF k-NN Decision Tree	Acc=97.30 Prec=97.73 Sens=97.04 F1=97.41 Spec=97.60
[36]	Private	304 brain MRIs	57-Normal MRIs 247-Brain Tumors	SVM Kalman Filter	Acc=98.02 Spec=95.39 Sens=97.04
[37]	Kaggle	7023 MR	1621-Glioma 1645-Meningioma 2000-No tumor 1757-Pituitary	CNN	Acc=97.15 Prec=0.97 Recall=0.97 F1=0.97
[38]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	SVM	Acc=99.15 Prec=0.97 Recall=0.97 F1=0.97

accuracy, demonstrating that carefully engineered features can rival or even surpass DL methods. On more diverse public datasets such as Figshare, ML performance typically falls within the 90–96% range, with KNN and SVM consistently emerging as the most reliable classifiers. Ensemble strategies such as RF–KNN–DT or SVM–ANN–KNN further enhance robustness by combining complementary decision boundaries, particularly in balanced or binary scenarios. These findings suggest that while ML methods may be limited by feature representation on heterogeneous datasets, they remain highly effective and competitive for smaller, cleaner, and more homogeneous MRI collections.

2) TUMOR CLASSIFICATION WITH DEEP LEARNING

Khan et al. [39] proposed a hierarchical DL method for brain tumor classification. In this study, the researchers used a Kaggle dataset consisting of MR images of gliomas, meningiomas, pituitary tumors, and no-tumor cases. They applied data resizing, data normalization, and data augmentation and achieved 92.13% classification accuracy. Ghassemi et al. [40] presented a new brain tumor classification model that leveraged pre-training as a discriminator within a generative adversarial network (GAN) to capture key features effectively. Subsequently, the final fully connected layer was

replaced with a SoftMax classifier to distinguish between three types of tumors. The proposed model comprises six layers and is employed alongside diverse data augmentation techniques. It attains an accuracy of 93.01% on the Figshare dataset. Amou et al. [41] proposed developed a deep CNN architecture for brain tumor classification. This architecture employs a pioneering technique based on Bayesian optimization to optimize the hyperparameters of CNNs. Isunuri and Kakarla [42] proposed a seven-layer CNN to address three types of brain tumor (glioma, meningioma and pituitary) MR image classification. They also optimized computation time through separable convolution. They achieved 97.42% accuracy in their study using Figshare dataset. Alhassan and Zainon [43] introduced an innovative deep Convolutional Neural Network (CNN) architecture utilizing the hard swish-based ReLU activation function for the classification of brain MRI into three classes: glioma, meningioma, and pituitary tumor. The incorporation of the hard swish-based ReLU activation function led to a notable enhancement in classification performance, achieving an increased accuracy of 3.5%, with the highest accuracy reaching 98.26%. In the study by Pashaei et al. [44], a CNN architecture was designed to extract features from brain MRI images. The features extracted by the CNN served as the input for the Kernel

Extreme Learning Machine (KELM) method. The KELM method is a learning algorithm that involves layers of hidden nodes for processing. Subsequently, the same CNN model was used for classification. The model consisted of five learnable layers, with a filter size of 3×3 in all layers. The CNN model achieved a classification accuracy of 81%. However, at the conclusion of this study, recall scores for the class 'pituitary tumor' were very high, while those for 'meningioma tumor' were notably low. This suggests limitations in the classifier's ability to discriminate between these classes [29]. Raza et al. [45] proposed a hybrid DL model called DeepTumorNet for three types of brain tumors. They used the GoogLeNet architecture as the base model. In the hybrid DeepTumorNet architecture, the final 5 layers of GoogLeNet were removed, and 15 new layers were incorporated. The proposed model obtained 99.67% accuracy, 99.6% precision, 100% recall, and a 99.66% F1-score on the Figshare dataset. Swati et al. [46] have implemented an approach for the classification of brain tumor MR images classification using pre-trained deep CNN (VGG19) transfer learning and fine-tuning. In this study, an accuracy of 94.82% was achieved under the five-fold cross-validation mechanism. Samee et al. [47] introduced a hybrid transfer learning architecture called GN-AlexNet, constructed by combining ten layers from AlexNet with five layers from GoogLeNet. The performance of this model was evaluated against several well-known pre-trained networks, including VGG-16, AlexNet, SqueezeNet, ResNet, and MobileNet V2. GN-AlexNet performed better than these models. The model's accuracy rate of was 99.51% Maqsood et al. [48] introduced a custom 17-layer deep neural network architecture for brain tumor segmentation. Then, they used a modified MobileNetV2 architecture for feature extraction and trained using transfer learning. They developed a Multiclass SVM (M-SVM) to divide MRIs into three tumor categories (glioma, meningioma, and pituitary). As a result of the study, they achieved an accuracy rate of 97.47% on the BraTs 2018 dataset and 98.92% on the Figshare dataset. Alanazi et al. [49] have developed a framework for brain tumor classification employing both isolated and transfer learning techniques. CNN models were created from the ground up to check performance. Then, using the transfer-learning technique, the weights of the neurons are changed and reused to classify the brain MRI images into tumor subclasses using binary classifier (tumor and no tumor) including a related CNN with 22-layer CNN model. Gómez et al. [50] conducted an evaluation of seven deep CNN models, comprising one generic CNN model and six pretrained models, for the multiclass classification of brain tumors, including gliomas, meningiomas, pituitary tumors, and non-tumorous brains. The authors used ResNet50, MobileNetV2, InceptionV3, InceptionResNetV2, Xception and EfficientNetB0 models in the transfer learning approach. They used the Msoud dataset, which includes Figshare, SARTAJ, and Br35H datasets. The

best result for this dataset was found to be InceptionV3, which reached an average accuracy rate of 97.12% of the CNN model. Ullah et al. [51] introduced a novel unified end-to-end DL model, named TumorDetNet, designed for brain tumor detection and classification. The TumorDetNet framework incorporates 48 convolutional layers to perform both the detection and classification of brain tumor into benign or malignant brain tumors and multiple tumor types (meningiomas, pituitary, and gliomas). Authors achieved 99.27% accuracy in their study using Kaggle data, also evaluated their model on the CE-MRI figshare dataset and achieved 98.47% accuracy. Ghosal et al. [52] proposed a DL framework based on the Squeeze-and-Excitation ResNet-101 model for automated tumor classification in MRI scans. Their work focused on three tumor categories—meningioma, glioma, and pituitary—and applied preprocessing steps such as zero-centering and intensity normalization. They also compared the assessment results using data augmented. The accuracy value was found to be 89.93%, and with the use of augmented data, this value has increased to 93.83%. Sajjad et al. [53] presented a CNN-based brain tumor type classification and multi-grade classification systems. Experiments were performed on two different datasets – Radiopaedia and CE-MRI. Firstly they segmented tumors using a DL approach. Then extensive augmentation and fine-tuned pre-trained model (with VGG-19) using data augmented. Alqudah et al. [54] put a custom DCNN architecture with 18 layers for classification of tumor types, also researched the effect of with use cropped, uncropped and segmented lesion images the experiment. They have obtained the accuracy 98.93%, 99% and 97.62% for cropped, uncropped and segmented images, respectively. Díaz-Pernas [55] proposed a multi-scale deep CNN capable of performing both segmentation and classification of brain tumors in MRI scans. The model was evaluated on the Figshare dataset, achieving a classification accuracy of 97.3%. Shaik and Cherukuri [56] developed a Multi-Level Attention Mechanism Network (MANet) aimed at identifying brain tumors. The MANet framework not only prioritized tumor locations but also preserved cross-channel temporal relationships within semantic feature sequences extracted from the Xception backbone. The model achieved an accuracy of 94.91% on the BraTS 2018 dataset and 96.51% on the Figshare dataset. Kibriya et al. [57] designed a completely automated brain tumor classification framework that uses transfer learned CNN models for deep feature extraction. They normalized image intensity and augmented dataset. GoogLeNet, AlexNet and ResNet18 deep CNN models were used for the transfer learning and create a single feature vector. SVM and KNN classifier is used on this fused feature vector and they obtained 97.7% accuracy. Rehman et al. [58] proposed using pre-trained CNN models (AlexNet, GoogleNet and VGG16) to classify the brain tumors. The authors performed contrast enhancement and augmented the dataset during the preprocessing phase. The evaluation findings indicated

that the fine-tuned VGG16 approach achieved the highest accuracy of 98.69%. Montoya et al. [59] employed both shallow and deep neural network methodologies to evaluate their efficacy in multiclass brain tumor classification. Their proposed deep network, based on the ResNet50 model and fine-tuning, demonstrated superior classification performance compared to Multi-Layer Perceptrons (MLP) and CNN models constructed from scratch. Deepak et al. used a pretrained GoogLeNet to extract features from brain MR images using a deep CNN, effectively differentiating between glioma, meningioma, and pituitary tumor types. The suggested system achieves an average classification accuracy of 98%. Bhanothu et al. [60] presented a region proposal network and faster R-CNN to brain tumor detection and classification. To ensure the efficient implementation of Faster R-CNN, they employed the VGG-16 architecture as the foundational mathematical model. Region Proposal Network was employed to precisely delineate the tumor area. They achieved 77.60% Mean Average Precision (MAP) score from their model. Kumar et al. [61] demonstrated the effectiveness of the CNN architecture (Alexnet, ResNet50, and Inception V3) for brain tumor identification using augmented MRI data. An accuracy of 98.24% is obtained for AlexNet at the 120th epoch with augmentation and performed better than other models. Nayak et al. [62] for classify the brain tumor into four categories (glioma, meningioma, pituitary and no tumor) developed a CNN model using min-max normalization. This model was derived from the EfficientNet architecture. The result accurate rate is 98.78%. Badža and Barjaktarović [63] presented a new 22-layered CNN architecture for brain tumor-type classification. they utilized the data-augmentation and 10—fold cross-validation. The authors achieved an accuracy of 96.56% for the multiclass brain tumor classification on the augmented dataset. Ismael and Abdel-Qader [23] suggested a classification algorithm based extracted statistical features from MRI slices with the aid of two-dimensional DWT and Gabor filter techniques. Researchers employed classification of brain tumor type with back-propagation neural network classifier and achieved an accuracy rate of 91.9%. Noreen et al. [64] used both DL and ML techniques for brain tumor classification. They proposed an ensemble method that uses fine-tuned networks, such as Inception-V3 and Xception, to extract features and integrated with combines various classifiers: Softmax, SVM, K-Nearest Neighbours, and Random Forest for classification. Proposed ensemble approach has classified the brain tumors with 94.34% overall accuracy by using InceptionV3. Deepak and Ameer [65] designed a lightweight CNN model consisting of five layers to extract image features from MRI images and used the SVM classifier on extracted features. They achieved 95.82% accuracy in brain tumor classification with this method. Nassar et al. [66] developed a hybrid classification approach in which the final decision among three brain tumor categories (meningioma, glioma, and pituitary) was determined through majority voting, combining outputs from

five pre-trained and fine-tuned CNNs: GoogleNet, AlexNet, ShuffleNet, SqueezeNet, and NASNet-Mobile. The proposed system prediction accuracy achieved 99.31%. Biswas and Islam [67] introduced K-means algorithm for preprocessing and artificial neural network (ANN) for the classification of three-class brain tumours. Discrete wavelet transform was used to extract features of segmented brain tumors with K-Means algorithm. Principal Component Analysis was employed to reduce the features. Then, ANN was developed using Levenberg-Marquardt function to classify Glioma, Meningioma and Pituitary tumors. Suggested approach provided 95.4% accuracy. Ismael et al. [68] used ResNet50 network to classify different brain tumors. They augmented the dataset and achieved an accuracy of 99% at the slice level and 97% at the patient level. Cinar et al. [69] presented a CNN architecture from scratch for classifying different types of brain tumors through MR images. They achieved accuracy 98.32% on average. Nawaz et al. [70] presented DenseNet-41-based CornerNet framework to classify brain tumors. They attained an average accuracy of 98.8% at Figshare dataset and 98.5% at Brain MRI dataset. Rahman and Islam [71] proposed a novel parallel deep convolutional neural network (PDCNN) topology for classification MR-based brain tumor types. They tested their work on 3 different data sets (binary tumor identification dataset-I, Figshare dataset-II and Multiclass Kaggle dataset-III). This topology is designed to extract both global and local features through parallel stages. The accuracy rates attained 97.3%, 97.6% and 98.1% on these datasets. Gaur et al. [72] used a CNN-based model to identify brain tumor subtypes. Also, for explanation prediction local interpretable model-agnostic explanation, and Shapley additive explanation were used. They added Gaussian noise to enhance the model's robustness to noise and improve its classification performance. This proposed method achieved an accuracy of 94.64%. Artzi et al. [73] introduced a fused architecture combining two neural networks—a pre-trained ResNet-50 and a tabular-based model—to classify posterior fossa tumors, including Pilocytic Astrocytoma, Medulloblastoma, and Ependymoma, as well as healthy brain cases. They achieved accuracy of 0.87. Noreen et al. [74] introduced multi-level features extraction and concatenation using pre-trained Inception-v3 and DensNet201 for detection of brain tumor types. Then, features were passed to softmax classifier. The approach had accuracy of 99.34% at Inception-v3 and 99.51% at DensNet201. McAvoy et al. [75] employed a CNN-based analytical approach to assist radiologists in making accurate differential diagnoses between primary CNS lymphoma (PCNSL) and glioblastoma multiforme (GBM). This approach yielded an AUC of 0.94 for GBM and an AUC of 0.95 for PCNSL with image augmentation. Ye et al. [76] used pretrained ResNet18 for classify Germinomas of the basal ganglia and gliomas. Brain lesions segmentation process was done manually in T1C and T2 FLAIR sequences. T1C sequences were trained with transfer learning approach.

This approach reached mean AUC of 88% and a mean accuracy of 81%. Anaraki et al. [77] presented a CNN and genetic algorithm (GA)-based method to classify glioma, meningioma and pituitary tumors. Rizwan et al. [78] focused on classifying the types of brain tumors and grades of glioma. They utilized Gaussian Convolutional Neural Network (GCNN) on the Figshare dataset to classify brain tumors types into pituitary, glioma, and meningioma. Their proposed approach achieved an accuracy of 99.8%. Hassan and Ghadiri [80] aimed to classify glioma, meningioma, pituitary tumors, and tumor-free cases using MRI images. For this purpose, they used the Msoud dataset. This dataset combines the Figshare, SARTAJ, and Br35H datasets. After data augmentation, they used the EfficientNetV2 model for transfer learning and feature extraction. They achieved a high accuracy of 99%. Hosny et al. [81] introduced an interpretable ensemble DL framework designed for brain tumor detection and classification based on MRI scans, employing a publicly available contrast-enhanced dataset from Figshare. The authors applied preprocessing steps including normalization, cropping to extract the region of interest (ROI), and resizing. Their ensemble model combined DenseNet121 and InceptionV3 pre-trained networks, where the classifiers were replaced with new layers tailored to the task. The models were fine-tuned via transfer learning and combined using a soft voting approach. Additionally, the Grad-CAM method was integrated to provide explainable visualizations, highlighting critical regions influencing the model's predictions. Experimental results demonstrated that the ensemble model outperformed individual networks and existing methods, achieving 99.02% accuracy, 98.75% precision, 98.98% recall, and 98.86% F1-score on the test set. The study highlights the potential of explainable ensemble learning in improving brain tumor classification performance and trustworthiness in medical applications. Asghari Ilani et al. [82] proposed a hybrid DL framework for brain tumor classification using T1-weighted MRI images. The study employed the publicly available Figshare dataset containing 3064 images of glioma, meningioma, and pituitary tumors. They compared a 3-layer CNN, U-Net, and several transfer learning models (VGG19, InceptionV3, EfficientNetB4). U-Net achieved the best performance with 98.56% accuracy, F1-score of 99%, and AUC of 99.8%, outperforming other models. The study also validated U-Net's robustness on an external dataset, where it achieved 96.01% accuracy. The results highlighted the effectiveness of U-Net and transfer learning approaches in brain tumor classification tasks. Qureshi et al. [83] proposed two enhanced DL architectures, SAlexNet-1 and SAlexNet-2, for multi-class brain tumor classification (glioma, meningioma, pituitary) as well as binary malignancy detection. Their approach integrates a Hybrid Attention Mechanism (HAM), dense feature extraction via multiple 3×3 convolution layers, and a semi-transfer learning strategy to improve representational richness. The models were evaluated on two large publicly available MRI datasets. SAlexNet-1 achieved accuracies

of 98.78% (multi-class) and 98.07% (binary), while the extended SAlexNet-2 achieved state-of-the-art accuracies of 99.69% and 99.17%, outperforming previous methods. Additionally, the authors incorporated Grad-CAM to visualize HAM activations, demonstrating the model's ability to focus on clinically relevant tumor regions, thereby increasing interpretability.

In summary, DL approaches achieve the highest performance on large and heterogeneous datasets, benefiting substantially from transfer learning and advanced convolutional architectures. Models built on EfficientNet, DenseNet, Inception, and ResNet routinely reach 97–99.7% accuracy, outperforming classical ML techniques on complex, multi-source MRI datasets such as Figshare, Kaggle, and Msoud. Ensemble CNN frameworks—where multiple pretrained networks are combined—push performance even further, exceeding 99% in several studies and representing the current state-of-the-art. Hybrid methods that integrate deep CNN features with classical classifiers such as SVM or KNN also show competitive performance, often closely approaching full DL ensembles. In contrast, shallow or custom CNNs tend to yield more variable results and are more sensitive to data diversity and augmentation strategies. Collectively, these findings indicate that transfer learning and ensemble-based DL models offer the most robust and generalizable solutions for MRI-based brain tumor classification.

Beyond tumor type identification, determining tumor grade is equally critical for treatment planning and prognosis, motivating the need for dedicated grade classification studies.

D. REVIEWS ON BRAIN TUMOR GRADING METHODS

Brain tumors are classified into grades based on their cellular characteristics, growth rates, and levels of malignancy. It is possible to determine brain tumor grades using MR imaging, one of the imaging methods. To accelerate and simplify this process, DL and ML methods are utilized. In Section III-C1, studies using ML for brain tumor grading are presented, while Section III-C2 focuses on studies employing DL methods.

1) TUMOR GRADE CLASSIFICATION WITH MACHINE LEARNING

Gupta et al. [84] devised a novel clinical decision support system that combines data from different MRI sequences and applies automatic threshold-based methods for precise image segmentation. They used the Naive Bayes classifier to classify glioma tumors into low-grade and high grade, and evaluated their method on two datasets. The authors achieved an accuracy of 97.83%. Citak et al. [85] used three distinct ML algorithms (SVM, MLP and logistic regression) to differentiate low-grade gliomas (WHO grades I and II) from high-grade gliomas (WHO grades III and IV). As a result, they achieved a 93% accuracy rate, along with 86.7% specificity and 96.4% sensitivity. Zhang et al. [86] applied various ML algorithms to classify gliomas to classify gliomas into LGG and HGG categories as well as

TABLE 3. Tumor classification with deep learning.

Paper No	Data			Detection Methods	Metrics
	Source	Patient Number	Tumor Type		
[39]	Kaggle	3264 images	926-Glioma 937-Meningioma 500-No tumor 901-Pituitary	CNN	Acc=92.13 Miss rate=7.87
[40]	Kaggle	3264 images	926-Glioma 937-Meningioma 500-No tumor 901-Pituitary	CNN	Acc=92.13 Miss rate=7.87
[41]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN-GAN	Acc=95.6 Prec=95.29 Sens=94.91 Spec=97.69 F1 score=95.10
[42]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=98.70 Prec=98.3 Sens=98.6 F1 score=98.6
[43]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=97.42 Prec=97.41 Sens=97.42
[44]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=98.60 Prec=99.6 Recall=98.6 F1 score=99.0
[29]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN ELM	Acc=93.68
[45]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	GoogLeNet	Acc=99.67 Prec=99.6 Recall=100 F1 score=99.66
[46]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	VGG19	Acc=94.82 Sens=94.25 Spec=94.69 Prec=89.52 F1 score=91.73
[47]	Figshare	262	760-Meningioma 1426-Glioma 940-Pituitary	GoogleNet AlexNet	Acc=99.51 Sens=98.90
[48]	BraTs 2018	No info	No info	SVM	Acc=97.47 Sens=97.22 Spec=97.94 Dice coefficient index=96.71
	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary		Acc=98.92
[49]	Kaggle	2780 images	826-Glioma 822-Meningioma 395-No tumor 827-Pituitary	CNN	Acc=95.75

TABLE 3. (Continued.) Tumor classification with deep learning.

	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary		Acc=96.89
[50]	Kaggle	7023 MR	1621-Glioma 1645-Meningioma 2000-No tumor 1757-Pituitary	CNN Inception-V	Acc=97.12 Prec=97.97 Recall=96.59 Spec=99.98 AUC=99.84
[51]	Kaggle	2764 images	926-Glioma 937-Meningioma 901-Pituitary	CNN	Acc=99.27 Prec=99 Recall=98.66 Spec=100 F-measure=98.82
	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary		Acc=98.47 Spec =96.73 Prec=98 Recall=97 F-measure=97.49
[52]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	Squeeze Excitation ResNet CNN	Acc=93.83 Spec=97.15 Sens=93.84
[53]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	VGG19	Acc=94.58 Sens=88.41 Spec=96.12
[54]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=98.93 Sens=98.18
[55]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	Multiscale CNN	Acc=97.30
[56]	Figshare	233 patients 843 images	300-Meningioma 343-Glioma 200-Pituitary	Xception	Acc=96.51
[57]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	SVM k-NN	Acc=99.7 Recall=100 Prec=99 F1-Score=99
[58]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	VGG16	Acc=98.69
[59]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	MLP CNN ResNet50	Acc=97.0
[60]	Figshare	233 2406 slices	805-Glioma 694-Meningioma 907-Pituitary	Faster R-CNN	mAP=77.6
[61]	Figshare	233 843 images	300-Meningioma 343-Glioma 200-Pituitary	AlexNet	Acc=98.24
[62]	Kaggle& Figshare	233 patients 3260 slices	708-Meningioma 1426-Glioma 930-Pituitary	EfficientNet	Acc=98.78 Prec=98.75 F1 score=98.75
[63]	Figshare	233	196-No Tumors 708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=96.56 Prec=95.79 Recall=96.51 F1 score=96.11

TABLE 3. (Continued.) Tumor classification with deep learning.

[64]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	k-NN SVM RF	Acc=94.34
[65]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN SVM	Acc=95.82
[66]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	GoogleNet AlexNet ShuffleNet SqueezeNet NASNetMobile	Acc=99.31 Sens=98.30 Spec=99.01 Prec=99.34 F1=98.90
[67]	Figshare	563	246-Glioma 74-Meningioma 243-Pituitary	K-means Clus- tering ANN	Acc=95.4 Sens=94.58 Spec=97.83
[68]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	ResNet50	Patient-level: Acc=97 Prec=98 Recall=97 F1-Score=97 Balanced Acc=97 Image-level: Acc=99 Prec=99 Recall=99 F1-Score=99 Balanced Acc=99
[69]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=98.32
[70]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary	DenseNet-41	Acc=98.8
[71]	Figshare	233 3064 MRI	708-Meningioma 1426-Glioma 930-Pituitary	CNN	Acc=97.60 Prec=97 Recall=97 F1 score=97
	Kaggle	2780 images	826-Glioma 822-Meningioma 395-No Tumor 827-Pituitary	CNN	Acc=98.12 Prec=98 Recall=97.75 F1 score=98
[72]	Kaggle	2870 images	Glioma Meningioma Pituitary	CNN	Acc=94.64
[73]	Private	158 MRI	22-Normal 63-Pilocytic AST 57-Medulloblastoma 16- Ependymoma	CNN	Acc=87.0
[74]	Figshare	233	708-Meningioma 1426-Glioma 930-Pituitary tumor	DenseNet201	Acc=99.51
[75]	Private	320 3887 images	160-GBM 160-PCNSL	EfficientNetB4	Acc=91.0 GBM: AUC=94 PCNSL: AUC=95

TABLE 3. (Continued.) Tumor classification with deep learning.

[76]	Private	73	32-Germinoma 41-Glioma	ResNet18	Acc=81.0 AUC=88
[77]	Figshare	233	708-Meningioma 1426-Glioma	Genetic Algorithm- CNN	Acc=94.2
[78]	Figshare	230 3061 slices	930-Pituitary 707-Meningioma 1425-Glioma	Gaussian CNN	Acc=99.8
[79]	Kaggle	7023 MR	929-Pituitary 1621-Glioma 1645-Meningioma	EfficientNetV2	Acc=99.16 Prec=99 Recall=99 F1-Score=99
[80]	Kaggle	10288 images	2000-No tumor 1757 Pituitary 2548-Glioma 2658-Pituitary 2582-Meningioma 2500-No tumor	YOLOv7	Acc=99.5 Prec=99.5 Recall=99.3 Sens=99.3 Spec=99.4 F1-Score=99.4
[81]	Figshare	230 3061 slices	707-Meningioma 1425-Glioma 929-Pituitary	Ensemble CNN	Acc=99.02 Prec=98.75 Recall=98.98 F1-Score=98.86
[82]	Figshare	230 3061 slices	707-Meningioma 1425-Glioma 929-Pituitary	U-Net	Acc=98.56 AUC=98.8 Recall=99 Prec=99 F1-Score=99
[83]	Kaggle	7023 MR	1621-Glioma 1645-Meningioma 1757-Pituitary 2000-No-tumor	SAlexNet-1/2	Acc=99.69 Prec=99 Recall=99 F1=99

WHO grades II, III, and IV. Their findings indicated that the SVM outperformed the other classifiers in terms of performance. An accuracy of 0.945 was obtained for LGG and HGG and an accuracy of 0.961 for grade II, III, and IV gliomas. Gupta et al. [22] performed a study to categorize gliomas into low-grade and high-grade classifications. They suggested a hybrid methodology for feature extraction from brain MRI. The tumor's boundaries were identified on 3D images using a combination of the DWT and local binary pattern (LBP) approaches. They obtained an accuracy of 96% with the Naïve Bayes classifier. Ranjith et al. [87] aimed to distinguish benign and malignant glioma tumors using MRI spectroscopy. Grade II gliomas were considered benign and Grade III and Grade IV gliomas were considered malignant. They used four different ML methods: MLP, SVM, random forest, and locally weighted learning. Using the random forest algorithm, the best AUC value was obtained at 0.911, while using locally weighted learning, the highest sensitivity rate was obtained at 86.1%. Ramdlon et al. [88] introduced a tumor classification system that detects tumors and edema in T1 axial contrast and T2 axial WI FSE image sequences and classifies tumor types into

three classes: Astrocytoma (AST), Glioblastoma (GLI), and Oligodendroglioma (OLI). They used basic image processing techniques in the segmentation stages. The KNN classifier was used to classify tumors. Hamerla et al. [89] used a ML approach to classify meningioma grades as I and high (II-III) grades. For this purpose, AUC maps, T1, T1-contrast, subtraction maps, FLAIR, and T2 MR images of 138 patients were acquired at five different centers. The segmentation step was done manually. They used different classification models, including Random Forest, XGBoost, SVM, and MLP. The authors achieved an AUC of 0.97 using an XGBoost classifier. Al-Saffar and Yildirim [90] developed a multiple eigenvalues selection (MES) to predict the grades of glioma tumors. They used a classification method based on both SVM and MLP.

Overall, the reviewed ML-based tumor grade classification studies demonstrate that traditional classifiers can achieve high and clinically meaningful performance when supported by informative MRI features and appropriate pre-processing. Across multiple datasets, Naïve Bayes and SVM consistently emerge as top-performing algorithms, reaching 94–97.8% accuracy for LGG–HGG differentiation and

TABLE 4. Tumor grade classification with machine learning.

Paper No	Data				Detection Methods	Metrics
	Source	Patient Number	Tumor Type	Tumor Grade		
[84]	JMCD BraTs	134 80	Glioma Glioma	770-Non-tumor 200-AST 130-Ependymoma 150-OLI 90-GLI 80-LGG 120-HGG	Naïve Bayes	Acc=97.83 Acc=96.47
[85]	Private	43	Glioma	15-LGG 28-HGG	SVM	Acc=93.0 Sens=96.4 Spec=86.7
[86]	Private	120	Glioma	28-LGG 92-HGG	SVM	Acc=94.5
[22]	BraTs 2018	50	Glioma	29-HGG 21-LGG	Naive Bayes	Acc=96.0 Sens=97 Spec=95
[87]	Private	28	Glioma	16-Grade I 12-Grade II–III	RF	Sens=80.6 Spec=85.7 AUC=91.1
[88]	TCIA	254 images	Glioma	84-AST 124-GLI 46-OLI	k-NN	Acc=89.50
[89]	Private	147	Meningioma	102-Grade I 45-Grade II–III	XGBOOST	AUC=97 Sens=100 Spec=97
[90]	TCIA	160	Glioma	489-HGG 489-LGG 489-Normal	SVM	Acc=91.02 Recall=86.52 Spec=94.26 Prec=87.07

maintaining strong sensitivity and specificity values. Studies employing hybrid feature extraction approaches—such as DWT, LBP, multi-sequence MRI inputs, or spectroscopy-derived biomarkers—further enhance the discriminative capacity of ML models, achieving accuracies of 96–97% in several works. Ensemble and tree-based methods, particularly XGBoost, also show robust generalization in multi-center settings, obtaining AUC values up to 0.97 for meningioma grade prediction. Even though simple methods such as k-NN perform worse in multi-class glioma grading, most ML models remain stable and reliable when the features are well-designed and the MRI data quality is high. Collectively, these findings indicate that ML remains a strong and effective framework for glioma and meningioma grade classification, particularly in structured, feature-rich diagnostic pipelines.

2) TUMOR GRADE CLASSIFICATION WITH DEEP LEARNING TECHNIQUES

Sajjad et al. [53] proposed a novel CNN for multi-grade brain tumor classification. They fine-tuned the pre-trained VGG-19 CNN model and augmented data. the softmax classifier was applied, achieving an overall accuracy of 90.67%. Shaik and Cherukuri [56] devised a multi-level attention network (MANet) for the classification of brain tumors and glioma grades. For glioma grade classification, they used BraTs 2018 dataset and pre-trained Xception as a backbone model

for initial representation learning. At the end of the study, they achieved 94.91% accuracy. Chen et al. [36] conducted a tumor grading study involving 643 patients. Firstly, they segmented tumor images with a modified U-Net CNN. They then used DenseNet architecture for classifying tumors into low-grade and high-grade meningiomas and achieved an AUC of 92% and an accuracy rate of 90%. Mzoughi et al. [91] a 3D CNN model was designed that leverages volumetric T1-Gado MRI sequences to categorize brain gliomas as either LGG or HGG. Their research examined the influence of preprocessing approaches, including intensity normalization and adaptive contrast enhancement, along with data augmentation methods, on classification accuracy. The proposed model achieved an overall accuracy of 96.49% on the validation dataset. Wodzinski et al. [92] aimed to classify meningioma grades using heterogeneous MRI images (different scanner manufacturers, protocols, etc.). They used pretrained ResNet-18 network and obtained an accuracy above 74%. Banzato et al. [93] proposed a DL model based on apparent diffusion coefficient (ADC) maps derived from diffusion-weighted imaging (DWI) to classify meningioma grades into Grade I and Grade II–III. The authors evaluated two CNN architectures, AlexNet and Inception V3, applied to both post-contrast T1-weighted (PCT1W) images and ADC maps. However, the models trained on PCT1W images yielded poor grading performance, whereas the Inception V3–ADC model achieved an AUC of 0.94, outperforming the AlexNet–ADC model

(AUC = 0.63). In contrast, the Inception V3–PCT1W and AlexNet–PCT1W models demonstrated substantially lower AUC values of 0.68 and 0.55, respectively. Vasantachart et al. [94] utilized a novel asymmetric 3D CNN architecture using two encoding paths based on the T1post and T2-FLAIR MRIs to determine Grade I and II meningiomas. This approach predicted grades with 90% accuracy. Tandel et al. [95] aimed to classify gliomas into low-grade and high-grade. They divided the REMBRANDT data into its internal dataset as T1W, T2W, and FLAIR MRI sequences and categorized HGG and LGG. They suggested combining five CNN models (AlexNet, VGG16, ResNet18, GoogleNet and ResNet50) with ensemble algorithm based on majority voting (MajVot). Using FLAIR data, the best classification performance was discovered using MajVot with an accuracy value of 98.88%. Khazaei et al. [96] used EfficientNetB0 pre-trained model to classify HGG and LGG tumors. As a result of the study, the classification accuracy was obtained as 98.87%. Ozcan et al. [97] developed a custom CNN for glioma grade prediction. They expanded data using a variety of data augmentation methods. They compared the results of the research with pretrained AlexNet, GoogLeNet and SqueezeNet and their model was found to be more successful with 97% accuracy rate. Naser and Deen [98] used U-net model for automatic brain tumor segmentation and pretrained VGG16 model for grading gliomas of brain tumors. They classified gliomas into grade II and grade III and achieved 89% accuracy. Ahmad [99] aimed to segment and classify HGG and LGG from MR images. For this reason, they used hybrid image segmentation techniques. Author classified tumors with back-propagation neural network (BPNN), CNN, and SVM. SVM and CNN achieved an accuracy of 88%. Zhuge et al. [101] used T1, T1-Gd, T2, and T2-FLAIR images with 2D and 3D CNN models for automatic distinguishing low-grade (Grades II and III) glioma (LGG) and high-grade (grade IV) glioma (HGG). As a result of the experiment, it was found that 3D medical image classification yielded better performance. Lo et al. [102] used transfer learning with AlexNet to classify into grade II, III, and IV gliomas. The proposed CAD system employing the transferred DCNN along with data augmentation, attained a mean accuracy of 97.9%. Gupta et al. [103] introduced a CAD approach for glioma grade classification that utilized distinct segmentation methods tailored to various MRI pulse sequences, combined texture feature fusion, and implemented an ensemble classifier (includes SVM, naïve Bayes and kNN classifiers) using two image sets (T1-w and T1-pc sequences and T2-w and FLAIR sequences). Yang et al. [104] used transfer learning with AlexNet and GoogLeNet to glioma grading as LGG and HGG using MR images. Pretrained GoogLeNet had a better performance than AlexNet model with an accuracy of 94.5%. Kabir Anaraki [77] presented a CNN and genetic algorithm (GA)-based method to classify grading of gliomas into Grade II, Grade III and Grade IV. They achieved an accuracy of 90.9%. Rizwan et al. [78]

focused on classifying the types of brain tumors and grades of glioma. They used GCNN on the REMBRANDT dataset for the task of classifying the grade of gliomas as Grade II-III-IV. Their proposed approach achieved an accuracy of 97.14%. Decuyper et al. [105] proposed using 3D CNN to predict glioma grades. For the segmentation, they used 3D U-Net on BraTS 2019 dataset. Then they trained 3D tumor region on a large dataset. They tested their approach on TCIA test data and a completely independent dataset. Isunuri and Kakarla [106] introduced an ensemble coupled convolution network (ECCN) for the classification of three grades of brain tumors. They used pre-trained EfficientNetB0 for feature extraction and introduced a coupled convolution network to enhance semantic features via dual-path convolutions. The REMBRANDT dataset was utilized, containing T1-weighted MR images of brain tumors categorized into Grade II, Grade III, and Grade IV. Their model achieved a high accuracy of 96.95%, outperforming EfficientNet variants. Moreover, they reported strong precision, recall, and Jaccard scores, indicating robust performance in brain tumor grade classification. Srinivasan et al. [107] developed three deep CNN models for brain tumor classification from MRI images. The first model achieved 99.53% accuracy for binary classification of tumor presence. The second model classified brain tumors into five types—normal, glioma, meningioma, pituitary, and metastatic—with 93.81% accuracy. The third model focused on glioma grading (Grade II, III, and IV) and reached 98.56% accuracy. The models were trained on large publicly available datasets (Figshare, REMBRANDT, TCGA-LGG, TCIA) and their hyperparameters were fine-tuned automatically via grid search optimization. The proposed models outperformed classical CNNs like AlexNet, VGG-16, ResNet101, DenseNet121, and GoogleNet in all classification tasks. Tandel et al. [108] proposed an ensemble DL method for three-class brain tumor grade classification, including high-grade glioma (HGG), low-grade glioma (LGG), and normal brain tissue, using MRI images from the publicly available REMBRANDT dataset. Their method combined seven DL models and seven ML models through majority voting and was evaluated using five-fold cross-validation. The proposed approach achieved 98.55% accuracy for the three-class classification task, outperforming individual models. Additionally, explainable AI (LIME) was applied to enhance model interpretability.

Overall, DL-based tumor grade classification studies demonstrate consistently strong performance across a broad range of MRI datasets and architectures, particularly when leveraging transfer learning, multi-scale CNNs, or volumetric 3D CNN models. Pretrained CNN backbones such as VGG-19, Xception, Inception-V3, EfficientNetB0, and AlexNet commonly achieve 90–98.9% accuracy, with the highest results generally obtained on large, well-annotated datasets such as BraTS and TCIA. Models incorporating 3D convolutions and ensemble strategies show notable advantages in low-grade versus high-grade glioma

TABLE 5. Tumor grade classification with deep learning.

Paper No	Data				Detection Methods	Metrics
	Source	Patient Number	Tumor Type	Tumor Grade		
[53]	Radiopaedia	121	Menigioma Gliomas Glioblastomas	36-Grade I 32-Grade II 25-Grade III 28-Grade IV	VGG-19	Acc=90.67
[56]	BraTs 2018	-	Glioma	210-HGG 75-LGG	Xception	Acc=94.91
[36]	Private	643	Menigioma	518-Low Grade 125-High Grade	U-Net CNN	Acc = 90.1 AUC=0.918
[91]	BraTs 2018	284	Glioma	209-HGG 75-LGG	3D CNN	Acc=96.49
[92]	Private	174	Menigioma	126-Benign 48-Atypical	CNN	Acc=74.0 Recall=74
[93]	Private	117	Menigioma	79-Grade I 38-Grade II-III	Inception-V3, AlexNet	AUC=0.94, Sens=97.4 Spec=91.1
[94]	Private	96	Menigioma	55-Grade I 41-Grade II	Asymmetric CNN	Acc=90.0 Sens=85 Spec=93
[95]	TCIA (REMBRANDT)	1430	Glioma	663-LGG 767-HGG	Ensemble CNN (AlexNet, VGG16, ResNet18, GoogleNet, ResNet50)	Acc=98.88 Sens=98.95 Spec=98.80 AUC=98.88
[96]	BRATS-2019	335	Glioma	259-HGG 76-LGG	EfficientNetB0	Acc=98.8
[97]	Private	104	Glioma	50-LGG 54-HGG	CNN	Acc=97.1 Sens=98 F1 score=97 Spec = 96.3 AUC=98.9
[98]	TCIA	108	Glioma	50-Grade II 58-Grade III	VGG16	Acc=89.0 Sens=87 Spec=92
[99]	MICCAI BraTs 2015	124 images	Glioma	99-HGG 25-LGG	BPNN, SVM, CNN	Acc=88.0
[100]	CPM-RadPath Challenge 2020	221	Glioma	133-GBM 54-AST 34-OLI	CNN	Acc=87.8 Sens=77.2 Spec=93.0 AUC=90.2
[101]	BraTs 2018	285	Glioma	210-HGG 75-LGG	3D CNN	Acc=97.0
[102]	TCIA	130	Glioma	(30-Grade II 43-Grade III 57-Grade IV	AlexNet	Acc=97.9

classification, with several studies reporting accuracies around 96–99%. In meningioma grading, performance varies

more widely depending on dataset heterogeneity, ranging from 74% accuracy with simple CNNs to 0.94–0.97 AUC

TABLE 5. (Continued.) Tumor grade classification with deep learning.

[103]	Private	570 images	Glioma	200-AST 130-Ependymoma 150-OLI	SVM k-NN Naïve Bayes	Acc=98.38
	BraTs	320 images	Glioma	90-GLI 80-LGG 120-HGG		Acc=97.37
[104]	Private	113	Glioma	52-LGG grade II-III 61-HGG)	GoogLeNet	Acc=94.5, AUC=96.8
[77]	REMBRANDT TCGA-GBM TCGA-LGG Private	130 199 299 60	Glioma	- 199-Glioblastoma Multiform 299-HGG	Genetic Algorithm CNN	Acc=90.90
[78]	REMBRANDT	513 70 slices	Glioma	- 204-Grade II 128-Grade III 181-Grade IV	Gaussian CNN	Acc=97.14
[105]	TCGA-GBM CGA-LGG LGG- 1p19qDeletion BraTS 2019 Private	164 121 141 202 110	Glioma	337-GBM 291-LGG	3D CNN	Acc=90.0 AUC=93.28 Sens=80.26 Spec=93.48
[106]	REMBRANDT	130 7414 slices	Glioma	3568-Grade II 1795-Grade III 2051 Grade IV	Ensenmble Coupled CNN	Acc= 96.95 Prec=96.87 Recall= 96.55
[107]	REMBRANDT	133 4720 im- ages	Glioma	1712-Grade II 1296-Grade III 1712-Grade IV	CNN	Acc= 98.56
[108]	REMBRANDT	130 4720 im- ages	Glioma	1041-Normal 484-LGG 631-HGG	DL Ensemble (MajVot)	Acc=98.55 Recall=98.29 Prec=98.01 F1 Score=98.15

using more advanced architectures and diffusion-derived imaging biomarkers. Despite differences in data sources, MRI sequences, and class distributions, most DL models demonstrate strong generalization when combined with appropriate preprocessing, data augmentation, and segmentation strategies, indicating that DL frameworks remain highly

effective and scalable solutions for multi-class and binary tumor grade prediction.

IV. DISCUSSION AND CONCLUSION

In this study, presents a systematic literature review on brain tumor type classification and tumor grade classification using

DL and ML methods is presented. Publications of the existing literature in the last 10 years were included in this study. Following a thorough review, current research gaps were identified, providing directions for future research in this field. When positioned within the broader research landscape, most ML- and DL-based brain tumor classification studies remain at an experimental proof-of-concept stage rather than being clinically deployable systems. Although many models report high accuracy, their real-world applicability is limited by dataset homogeneity, lack of multimodal MRI integration, scarce external validation, and minimal attention to model interpretability. Despite strong results in controlled settings, these models still require rigorous multi-center validation and clinically aligned evaluation before they can be used in real clinical practice. Across the 81 reviewed studies, only a small subset incorporated interpretability analysis, indicating that explainability remains one of the least addressed aspects of current brain tumor classification research. Techniques such as Grad-CAM, SHAP, or counterfactual explanations can increase transparency by revealing which regions or features influence model decisions; however, their usage is inconsistent and often superficial. This lack of explainability further limits the clinical acceptance of current ML and DL systems.

Across the reviewed studies, several methodological limitations consistently emerge. First, many ML- and DL-based grading and classification models rely on small, single-center, or highly homogeneous datasets, which limits generalizability, especially for meningioma grading where dataset sizes were often below 150 patients. Second, most studies use inconsistent and minimal MRI modalities, frequently depending on only T1ce, T2-FLAIR, or ADC maps, despite the fact that real clinical diagnosis is inherently multimodal; only a few works integrate multiple sequences or assess their complementary value. Third, segmentation quality varies widely, with some studies using fully automatic pipelines (e.g., modified U-Net), while others rely on manual or partially manual tumor delineation—introducing observer bias and making comparisons across studies difficult. Fourth, although DL models often report very high accuracy (96–99%), many depend on simple architectures or transfer learning applied to 2D slices, which may lose important volumetric tumor information; even 3D CNNs often require aggressive preprocessing or augmentation to compensate for data scarcity. Fifth, class imbalance is a persistent issue—especially in grade II–III meningiomas and LGG vs. HGG gliomas—yet a large portion of the studies do not implement systematic imbalance handling (e.g., resampling, cost-sensitive learning), creating inflated accuracy for majority classes. Finally, performance metrics are inconsistently reported across works, and many papers emphasize accuracy while underreporting clinically critical measures such as sensitivity, specificity, or AUC. In clinical settings, evaluation metrics such as sensitivity, specificity, and F1-score play a crucial role in determining the practical utility of AI-based diagnostic systems. High sensitivity is

essential to minimize false-negative tumor predictions, while specificity is important to reduce false positives that may lead to unnecessary interventions. The F1-score provides a balanced assessment in cases of class imbalance, making it particularly valuable when minority tumor types or grades must be detected reliably. Therefore, these metrics directly shape the clinical decision-making value and safety of ML- and DL-based models. These limitations indicate that systematic imbalance-handling strategies (e.g., resampling or cost-sensitive learning), multimodal fusion, and the adoption of 3D architectures could help mitigate the generalizability issues observed in current studies. Collectively, these limitations highlight the need for larger multimodal datasets, standardized evaluation frameworks, and improved interpretability for ML and DL models to be reliably deployed in clinical practice.

Studies using Figshare and BraTs datasets are abundant. In addition, the sample sizes used in these experiments are often small. However, it is important to note that frequently used datasets such as Figshare and BraTS have several limitations that could affect the generalizability and clinical relevance of developed models. First, class imbalance is a common issue in these datasets, with certain tumor types or grades significantly underrepresented. Class imbalance often causes models to become biased, achieving good results for majority categories but failing to generalize effectively to underrepresented ones. Additionally, high accuracy values reported in some studies may be misleading in the presence of class imbalance, as accuracy tends to favor the majority class and may not reflect true model performance. Furthermore, most of these datasets are derived from limited sources and lack diversity in terms of imaging protocols, scanner types, and patient demographics. This limitation restricts the ability of models to generalize across different clinical environments. To address these challenges, future studies should prioritize the inclusion of multi-center, multi-device datasets and evaluate models on external data from diverse clinical settings to ensure robust and clinically relevant performance. Additionally, the number of data used in the experiments is small. For this reason, studies with a superior number of data stand out more.

In recent years, DL methods, especially transfer learning methods, have become increasingly popular in brain tumor type classification, whereas ML methods are less commonly explored. ML techniques, such as Support Vector Machines (SVM), Random Forest (RF), and k-Nearest Neighbors (k-NN), generally require smaller datasets and rely on manually extracted features. These approaches are often more interpretable and computationally efficient, making them suitable for studies with limited data or lower computational resources. However, their performance tends to plateau with larger, more complex datasets. In contrast, DL methods, particularly convolutional neural networks (CNNs), automatically learn hierarchical feature representations from raw images and often achieve superior accuracy, especially with large datasets. Nonetheless, DL models typically demand

higher computational resources and are criticized for their “black-box” nature, where the decision-making process is not easily interpretable. In addition to standalone ML and DL approaches, hybrid models that combine handcrafted radiomic features with deep representations have shown promising results in several studies. Such models leverage the interpretability and low-data robustness of ML techniques together with the hierarchical feature extraction capabilities of DL architectures. As reported in multiple works, these hybrid or feature-fusion strategies can improve performance, especially in scenarios with limited datasets, imbalanced classes, or heterogeneous imaging conditions. Therefore, incorporating and further exploring hybrid pipelines represents a valuable future research direction for strengthening tumor classification and grading models.

To address this limitation, explainable AI (XAI) techniques have gained attention in recent years. Methods such as saliency maps (e.g., Grad-CAM), feature attribution, and counterfactual explanations aim to enhance the transparency of DL models by visualizing the specific image regions or features that influence predictions. These tools help improve the trustworthiness of DL models and support their integration into clinical workflows by enabling healthcare professionals to verify whether the models focus on medically relevant areas. While XAI methods hold promise, they are not yet foolproof; some explanations may lack consistency or reliability. Nevertheless, integrating XAI with DL models represents a significant step toward making brain tumor classification systems more transparent, interpretable, and clinically acceptable. In clinical environments, XAI methods such as Grad-CAM can help clinicians assess whether the model focuses on medically meaningful tumor regions, thereby increasing trust in automated decisions. When the highlighted activation areas align with radiological findings, clinicians are more likely to rely on the predictions, whereas inconsistent or clinically irrelevant heatmaps may signal potential model failure. Thus, XAI tools not only improve transparency but also play a critical role in evaluating the clinical reliability and safety of AI-driven diagnostic systems.

Glioma has been used more frequently in brain tumor grade classification studies than other tumor types. One of the most important reasons for this is the availability of open data sources. Therefore, it can be said that there is a gap in the literature in studies based on grade predictions of tumors other than glioma. This represents a notable gap, particularly for tumor types with limited publicly available datasets. Moreover, unlike gliomas, many other intracranial tumor types—such as pituitary adenomas, schwannomas, craniopharyngiomas, and medulloblastomas—do not use WHO grade as a primary clinical determinant. Although meningioma grades I–III differ significantly in recurrence risk and therapeutic strategy, yet multi-class grading datasets remain scarce, leading many studies to merge grades or reduce the problem to a binary classification. As a result, grade-based classification studies are naturally concentrated on gliomas and, to a lesser extent, meningiomas, which

explains their overrepresentation in the literature and the limited availability of grade-labeled datasets for other tumor types. It has been observed that, due to limited data, tumor grades are often merged to simplify the classification task. As a result, multi-class grade classification (e.g., Grade I–II–III) remains an important open research problem for future studies.

Researchers frequently used ML and DL methods to detect brain tumor type and automatically classify their grades. A total of 81 articles were examined in this review. Each was briefly summarized and its important parameters reported. The use of methods such as transfer learning and ensemble techniques was associated with improved performance. In addition, it is understood from the literature that preprocessing methods are among the critical techniques for classification with MR images. Upon examining the articles, DL methods were found to be particularly successful in MRI classification and showed promise for tumor classification tasks. For instance, several of the reviewed studies reported high classification performance with DL models, often achieving accuracy levels above 95% and F1-scores exceeding 0.90 for tasks such as glioma grade classification or multi-class tumor type prediction. Overall, DL methods have shown strong potential for MRI-based classification and remain promising tools for future tumor classification and grading research.

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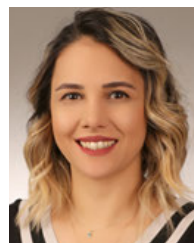
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